10-6 Data Analysis [5]

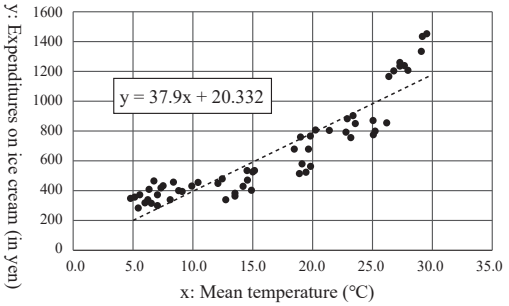
Point!

Regression Analysis

(1) ([1]**Regression analysis**) : A method to investigate and clarify the relationship between a resulting value and a causal value. It is used in predictions and simulations.

(2) ([2]**Simple regression analysis**) : Predicts the outcome variable y using the causal variable x.

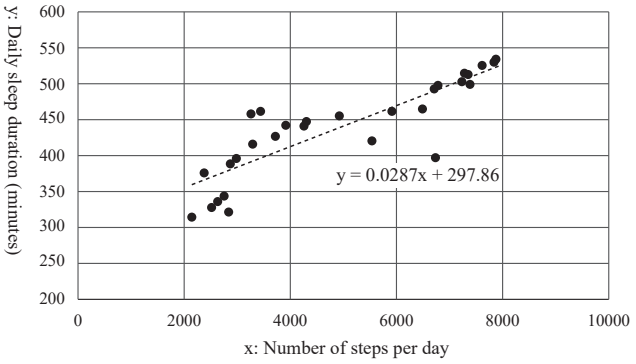
(3) ([3]**Regression line**) : Refers to the line used to determine predicted values in a scatter plot. A regression line can be expressed with the ([4]**regression line equation**) in the form of a first-degree function y = ax + b.



(4) ([5]**Least squares method**) : A method used to ([6]**minimize**) the vertical error, or the ([7]**residual**), between the actual data points and the regression line.

Warm Up

The following scatter plot summarizes the relationship between the number of steps and the sleep duration for a student over the course of one month. Answer the following questions about this graph.



(1) What is the term for describing the process of determining the equation for a straight line in the form y = ax + b representing the relationship between two variables, explanatory variable x and resulting variable y?

(2) Complete the following sentences by filling in the blanks [1] to [3] with the appropriate items.

> When representing the relationship between two variables with a straight line, the ([1]) method is for finding the line that best fits the actual data. In the ([1]) method, a straight line is determined to ([2]) the error, referred to as the ([3]), between each data point and the line.

(3) The straight line in the figure represents the relationship between two variables using a straight line. If the equation of this line is "y = 0.0287x + 297.86," what is the estimated sleep duration in minutes when the number of steps per day is 5,000? Give the answer as an integer (rounded off to the nearest whole number).

Explanation

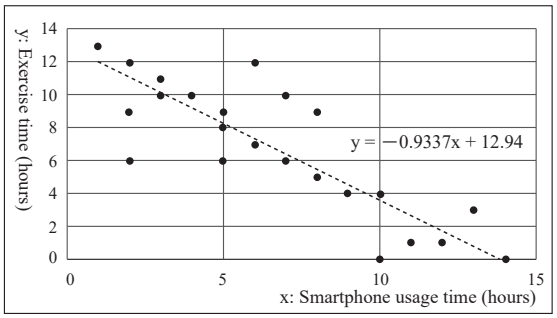
(1) Simple regression analysis

(2) [1] least squares [2] residual [3] minimize

(3) Substituting x = 5,000 into y = 0.0287x + 297.86, we get
y = 0.0287 × 5,000 + 297.86 = 441.36 ≈ 441
Therefore, the sleep duration is estimated at 441 minutes.

Try

The following figure is a scatter plot summarizing the results of a survey taken by 30 students from a class regarding their smartphone usage time and exercise time per day. Refer to this graph and answer the following questions.



(1) What is the term for the equation for a straight line in the form y = ax + b representing the relationship between two variables: causal variable x and resulting variable y?

(2) The straight line in the figure represents the relationship between two variables using a straight line. If the equation for this line is "y = − 0.9337x + 12.94," what is the estimated exercise time when smartphone usage is 10 hours? Give the answer as an integer (rounded off to the nearest whole number).

(3) Choose the term or number that best fits into the blanks [1] and [2] in the following sentences from the options **A** to **D** below.

> The least squares method is used to minimize the error between a straight line representing the relationship between two variables and the actual data. This type of error is referred to as ([1]). If the absolute value of the correlation coefficient is ([2]), the value obtained from the straight line determined by the least squares method is considered an estimate.

A Residual **B** Deviation **C** 1 **D** Other than 1

Exercise

1 The following figure is a scatter plot summarizing the measurement results of the height and weight of 100 adults. Refer to this graph and answer the following questions.

(1) Choose the one that best fits into the blanks [1] and [2] in the following sentences from the options **A** to **D** below.

> Based on the scatter plot, there is a tendency for weight to increase as height increases. Accordingly, when ([1]), it is referred to as a ([2]) between the two variables.

A Positive correlation
B Negative correlation
C One value increases and the other value also increases
D One value increases and the other value decreases

(2) The straight line in the figure represents the relationship between two variables using a straight line. If the equation for the line is "y = 0.5511x − 32.958," what is the approximate weight when the height is 173 cm? Give the answer as an integer (rounded off to the nearest whole number).